<table>
<tr><th colspan="7">B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level</th></tr>
<tr><td colspan="2">Sem:</td><td colspan="5">I</td></tr>
<tr><td colspan="2">Paper Category:</td><td colspan="5">DSC3-1 (Major) (Group - I)</td></tr>
<tr><td colspan="2">Paper Name:</td><td colspan="5">Practical based on DSC3-1 Paper Code : G06-0103-P</td></tr>
<tr><td colspan="2">Credit:</td><td colspan="2">02</td><td colspan="2">Practical:</td><td>4 Hrs./Week</td></tr>
<tr><td>Marks:</td><td>UA:</td><td>30</td><td>CA:</td><td>20</td><td>Total:</td><td>50</td></tr>
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Tools / Software: NASM (Netwide Assembler)/ MASM (Microsoft Macro Assembler) / TASM (Turbo Assembler)/ proteus etc.

1. Testing and implementation of Basic gates OR 7432, AND 7408, NOT 7404
2. Testing and implementation of universal gates NAND 7400, NOR 7402
3. Testing and implementation of EX-OR 7486 and EX-NOR 7486
4. Implementation of the basic gate using universal gate 7400 NAND gate
5. Implementation of the basic gate using universal gate 7402 NOR
6. Implementation of half adder using the basic gate
7. Implementation of half subtractor using the basic gate
8. Implementation of full adder using basic gate
9. Implementation of RS flip flop using NAND gate
10. Implementation of RS flip flop using NOR
11. Implementation of D-Flip Flop using IC 7474.
12. To write and execute ALP for the Addition of two, three, or multiple-byte numbers.
13. To write and execute ALP for Subtraction of two, three, or multiple-byte numbers.
14. To write and execute ALP for Multiplication of two, three, or multiple byte numbers.
15. To write and execute ALP for Division of 8 bit, 16 bit.
16. To write and execute assembly language program to arrange numbers in ascending order
17. To write and execute assembly language program to arrange numbers in

descending order

18. To write and execute assembly language program to transfer blocks of multiple byte number
19. To write and execute assembly language program for sorting of numbers.
20. To write and execute assembly language program for square of numbers.